

UniCal Codebase Guide

How to Ask ChatGPT for Help With Code Changes

Prepared for Bryn Kai-Hendricks — April 2026

1. Understanding Your Project Structure

UniCal is a full-stack JavaScript application with these key areas:

[client/src/pages/dashboard.tsx](#)

~28,000+ lines — The main dashboard with calendar, homework panel, weather, countdown bars, D2L ticker, Spotify, and everything you see on the home screen. This is by far the largest file.

[client/src/pages/files.tsx](#)

The OneDrive file browser / PDF reader / audio player page.

[client/src/pages/projects.tsx](#)

The project management page.

[server/routes.ts](#)

~21,000+ lines — All backend API endpoints (tasks, weather, calendar sync, OneDrive, email, essay templates, etc).

[server/storage.ts](#)

Database operations interface — all CRUD queries.

[server/index.ts](#)

Server startup, cron jobs, and middleware.

[server/onedrive.ts](#)

OneDrive API integration for file browsing and uploads.

[client/src/index.css](#)

Global CSS styles, theme variables, and custom classes.

[shared/schema.ts](#)

Database schema definitions and TypeScript types.

2. How to Describe What You Want to Change

Since your codebase is too large to upload entirely, you need to tell ChatGPT exactly where to look. Here is a template you can copy and paste:

```
I'm working on a full-stack JS app called UniCal.  
The frontend is React (Vite) and the backend is Express.  
  
I want to change: [DESCRIBE THE FEATURE OR BUG]  
  
The relevant code is in: [FILE PATH from the list below]  
Starting around line: [LINE NUMBER or "search for X"]  
  
Here is the specific code section I want to modify:  
```\n[PASTE the 10-50 lines of relevant code here]\n```\n  
What I want it to do instead:
[DESCRIBE THE DESIRED BEHAVIOR]
```

## 3. How to Find the Right Code to Paste

### Method A: Search by keyword on your Pi

SSH into your Raspberry Pi and use grep to search:

```
ssh pi@172.24.1.204
cd ~/Home-View
grep -n "countdown-bar" client/src/pages/dashboard.tsx | head -20
```

This shows line numbers where that text appears. Then extract the section:

```
sed -n '28280,28310p' client/src/pages/dashboard.tsx
```

### Method B: Search in VS Code

1. Open the project folder in VS Code (locally or via Remote SSH to Pi).
2. Press Ctrl+Shift+F to search across all files.
3. Type a keyword related to what you want to change (e.g., "white box", "hover", "countdown").
4. Click the result to jump to the exact line.
5. Select the relevant 10-50 lines and copy them.

### Method C: Use GitHub

1. Go to [github.com/Reachushere/Home-View](https://github.com/Reachushere/Home-View)
2. Click on the file you want (e.g., `client/src/pages/dashboard.tsx`).
3. Press "t" to search files by name, or use the search bar with "in this repository".
4. Click on a line number to highlight it. Shift+click another line to select a range.
5. Copy the URL — it now includes the line range (e.g., `#L28280-L28310`). Paste this URL into ChatGPT.

## 4. Common Change Scenarios & Where to Look

### "Change how a calendar event looks"

! dashboard.tsx — search for "gcal-event" or "CalendarEvent"

### "Change the homework panel"

! dashboard.tsx — search for "homework" or "HomeworkPanel"

### "Change weather display"

! dashboard.tsx — search for "weather" or "forecast"

### "Change countdown bars"

! dashboard.tsx — search for "countdown-bar" or "cbar"

### "Change the D2L ticker"

! dashboard.tsx — search for "d2l" or "ticker"

### "Change Spotify player"

! dashboard.tsx — search for "spotify" or "SpotifyPlayer"

### "Change file browser / PDF reader"

! files.tsx — search for what you want to change

### "Change project management"

! projects.tsx — search for what you want to change

### "Change a backend API endpoint"

! server/routes.ts — search for the API path (e.g., "/api/tasks")

### "Change what data is stored"

! shared/schema.ts for the DB schema, server/storage.ts for queries

### "Change global colors or CSS"

! client/src/index.css

### "Change OneDrive integration"

! server/onedrive.ts

### "Add or change a page route"

! client/src/App.tsx

## 5. Example Conversations with ChatGPT

### Example 1: Changing a color

You: "I have a React app. In dashboard.tsx around line 28290, there's a div with class cbar-hover-bg that has background: rgba(255,255,255,0.92). I want to change it to a light blue. Here's the code:"

```
<div className="cbar-hover-bg" style={{ background: 'rgba(255,255,255,0.92)' }} />
```

### Example 2: Adding a feature

You: "I have a full-stack JS app (React frontend, Express backend). I want to add a button that exports my tasks as a CSV. The tasks API is at /api/tasks in server/routes.ts. Here is the current task list component code from dashboard.tsx:"

[Then paste the 20-30 lines of the task list section]

### Example 3: Fixing a bug

You: "When I hover over a countdown bar, the white background box behind it isn't showing. The hover handler is in dashboard.tsx around line 28289. Here's the onMouseEnter code:"

[Then paste the relevant handler code]

## 6. Tips for Best Results

1. Always include the file name and approximate line number. ChatGPT can't guess where code lives.
2. Paste the actual code, not a description of it. "The hover handler" is vague; the actual 15 lines of code is precise.
3. Include surrounding context. Paste 5-10 extra lines above and below the target area so ChatGPT understands the structure.
4. Mention the tech stack upfront: "React + Express, TypeScript, Tailwind CSS, shadcn/ui, TanStack Query."
5. For styling changes, mention if you use inline styles (`style={{ }}`) or Tailwind classes (`className="..."`).
6. If ChatGPT gives you code back, it will usually show the exact replacement. You can use your text editor's Find & Replace to locate the old code and swap it in.
7. For large files (dashboard.tsx is ~28,000 lines), always give line numbers. Say "around line 5000" rather than just "somewhere in the file."
8. If you need to change something but don't know where it is, ask ChatGPT: "In a React app, where would I find the code that controls [X]?" — then search your codebase for the keywords it suggests.
9. You can paste a GitHub permalink URL (with line numbers) instead of pasting code directly. ChatGPT can read GitHub URLs.
10. Keep changes small and focused. One change per conversation works better than trying to do five things at once.

## 7. Quick Reference Card

```
FIND CODE: grep -n "keyword" client/src/pages/dashboard.tsx
EXTRACT CODE: sed -n '100,150p' client/src/pages/dashboard.tsx
VIEW ON WEB: github.com/Reachushere/Home-View !' navigate to file

TECH STACK: React + Vite | Express | TypeScript | Tailwind CSS
 shadcn/ui | TanStack Query | Drizzle ORM | PostgreSQL

BIG FILES: dashboard.tsx (~28,000 lines) – frontend UI
 server/routes.ts (~21,000 lines) – backend APIs

APPLY CHANGES: Edit the file on Pi, then restart:
 cd ~/Home-View && npm run dev
```